Teachertraining

Using CTFs in Cybersecurity Classes

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Workshop Agenda

1. Introduction & Discussion
2. What Are CTFs?
3. Tools for CTFs
4. CTF Platforms
5. Hands-On with CyberLeague
6. Wrap-Up & Resources

Introduction & Discussion

Warm-Up Question

Do you already use CTFs in your teaching?

If yes, how?

Let’s collect:

- Current methods
- Pain points
- Expectations for today

What Are CTFs?

CTF Definition

CTF

Capture the Flag (CTF) is a special kind of information security competitions. CTF games often touch on many aspects of information security: cryptography, stego, binary analysis, reverse engeneering, mobile security and others.

CTF Types

Jeopardy Style

- Individual challenges
- Categories: Web, Crypto, Pwn, Forensics
- Points-based scoring
- Most common in education

Attack/Defense

- Team-based
- Maintain services while attacking others
- Real-time competition
- Advanced level

Learning Benefits

- **Active Learning**
  - Hands-on problem solving
  - Learning by doing
- **Competency-Based Teaching**
  - Clear skill demonstration
  - Measurable progress
- **Discovery-Driven Learning**
  - Exploration and experimentation
  - Creative thinking

Gamification Elements

Why students engage with CTFs:

- 🎯 **Clear Goals** - Find the flag, earn points
- 📊 **Immediate Feedback** - Right or wrong, instant response
- 🏆 **Competition & Collaboration** - Scoreboards and teamwork
- 🔄 **Safe Failure** - Experiment without real-world consequences
- ⚡ **Motivation** - Challenging, rewarding, progressive

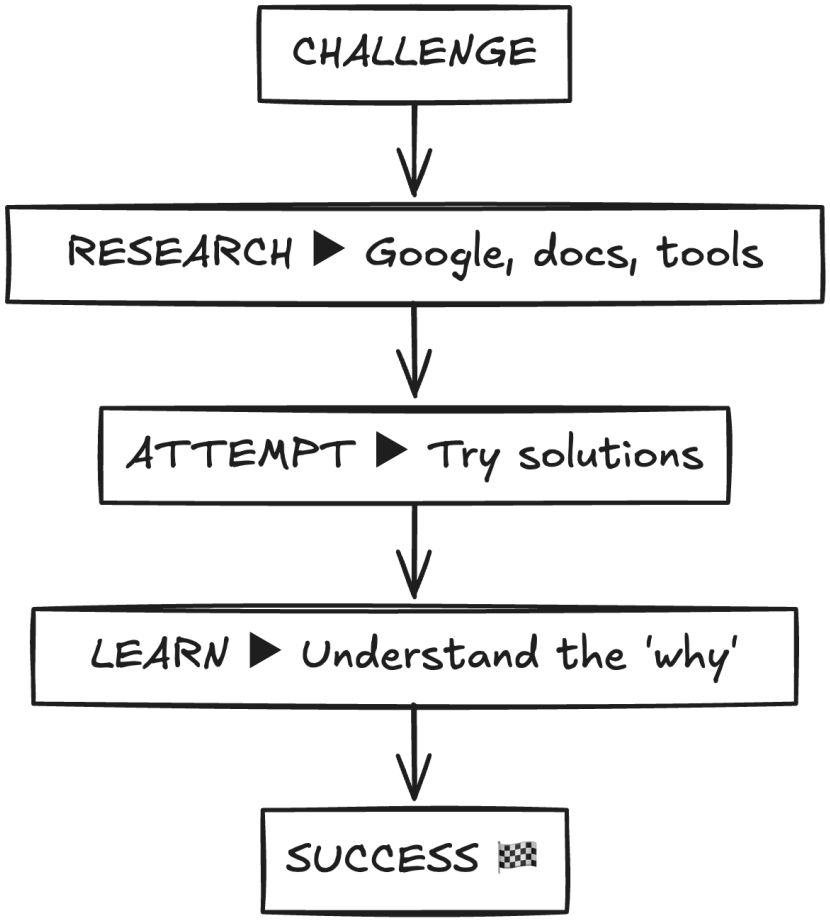
CTFs in Cybersecurity Education

Why CTFs belong in the classroom:

- Bridge theory and practice
- Develop problem-solving skills
- Build technical toolbox
- Encourage autonomous learning
- Simulate real-world scenarios
- Measurable learning outcomes

CTFs transform passive learners into active security practitioners

CTF Learning Cycle



Tools Every CTF Player Should Know

Networking & Basic Tools

Essential network utilities:

- **netcat** - Network Swiss Army knife
- **nmap** - Network scanner
- **Wireshark** - Packet analyzer
- **curl** - HTTP client

Example: Netcat basics

```
# Listen on port 1234
nc -lvnp 1234

# Connect to a service
nc example.com 80

# Send HTTP request
```

Networking & Basic Tools

```
8  echo -e "GET / HTTP/1.0\r\n\r\n" | nc example.com 80
```

Web / AppSec Tools

Web security toolkit:

- **Burp Suite** - Intercept and modify HTTP traffic
- **Browser DevTools** - Inspect, debug, modify
- **Postman / HTTPie** - API testing
- **SQLMap** - Automated SQL injection

Example: Burp Suite workflow

```
1. Configure browser proxy (127.0.0.1:8080)
2. Intercept requests
3. Modify parameters, headers, cookies
4. Observe responses
5. Use Repeater for testing
```

Reverse Engineering Tools

Binary analysis tools:

Disassemblers:

- Ghidra (NSA, free)
- IDA Free
- radare2 / Cutter

Basic utilities:

- `strings` - Extract text
- `file` - Identify file type
- `ltrace` / `strace` - Trace calls

Example: Basic analysis

```
# Check file type
file binary

# Extract strings
strings binary | grep flag

# Run in Ghidra
ghidra binary
```

Crypto & Misc Tools

Cryptography and general tools:

- **CyberChef** - The cyber Swiss Army knife (web-based)
- **Hashcat** - Password cracking
- **John the Ripper** - Hash cracking
- **Python** - Scripting everything

Scripting with Python

Example: Python brute force pattern

```python
import requests

for pin in range(10000):
    payload = {'pin': f'{pin:04d}'}
    r = requests.post('http://target.com/check', data=payload)

    if 'success' in r.text:
        print(f'Found: {pin:04d}')
        break
```

CyberChef Example

CyberChef - Decode, decode, decode

Common recipes:

- Base64 decode
- ROT13 / Caesar cipher
- XOR brute force
- Extract files from data
- Analyze hashes

🔗 **gchq.github.io/CyberChef**

CTF Platforms for Education

Popular Public Platforms

Global platforms:

- **picoCTF** – Beginner-friendly, CMU-hosted
- **HackTheBox** – Realistic machines, certification paths
- **TryHackMe** – Guided learning paths, very gamified
- **pwn.college** – Academic, ASU-hosted

Austrian & Regional:

- CyberLeague 24/7 (ctf.cyberleague.at)
- NodeZero SchoolCTF
- Coming Soon: NodeZero Cyberskills
- FuzzyLand (fuzzy.land)

Choosing the Right Platform

Consider:

- 📚 **Curriculum alignment** – Does it match your topics?
- 👥 **Student level** – Beginner, intermediate, advanced?
- 🎓 **Teacher tools** – Progress tracking, class management?
- 💰 **Cost** – Free, freemium, or paid?
- 🌍 **Language** – English, German, localized?
- 🔒 **Privacy** – GDPR, student data protection?

Hands-On with CyberLeague

CyberLeague Overview

What is CyberLeague?

- Austrian CTF platform
- Hosted at **ctf.cyberleague.at**

Getting Started

Step-by-step:

1. Visit **ctf.cyberleague.at**
2. Create account
3. Browse challenge categories
4. Select a challenge
5. Read description carefully
6. Submit flag
7. Check scoreboard

Let’s do this together now!

Challenge Categories

Typical CyberLeague categories:

- **Web** – XSS, SQLi, CSRF
- **Crypto** – Classic ciphers, modern crypto
- **Pwn** – Buffer overflows, exploits
- **Reverse Engineering** – Disassembly, patching
- **Forensics** – File analysis, memory dumps
- **Misc** – Logic puzzles, OSINT
- **Networking** – Packet analysis, protocols
- **Steganography** – Hidden messages

Recommended start: Web, Crypto, Misc (usually easier entry points)

Teacher Use Cases

1. **Weekly Assignments**
   - Assign 2-3 challenges per week
   - Match curriculum topics (Web security week → Web challenges)
2. **Class Competitions**
   - Host internal CTF events
   - Group or individual scoring
3. **Progress Tracking**
   - Identify struggling students early
4. **Extra Credit**
   - Bonus points for challenge solves
   - Motivates self-directed learning

Live Demo Time

Let’s solve a challenge together!



Instructor-led walkthrough

Wrap-Up & Resources

Benefits Summary

Why integrate CTFs into your teaching?

✅ **Engagement** - Students love the gamification

✅ **Skill Development** - Practical, hands-on learning

✅ **Autonomous Learning** - Self-paced exploration

✅ **Real-World Relevance** - Industry-aligned skills

✅ **Assessment** - Clear, measurable outcomes

✅ **Scalability** - Works for 10 or 100 students

Best Practices Checklist

For teachers implementing CTFs:

- [ ] Start with beginner-friendly platforms (picoCTF, CyberLeague)
- [ ] Align challenges with curriculum topics
- [ ] Provide time in class for exploration
- [ ] Offer hints but encourage struggle
- [ ] Celebrate first solves and creative solutions
- [ ] Mix individual and team challenges
- [ ] Debrief challenges: discuss methods and tools
- [ ] Gradually increase difficulty
- [ ] Monitor student progress and frustration levels
- [ ] Connect CTF skills to real-world scenarios

Common Pitfalls to Avoid

⚠️ **Watch out for:**

- **Too difficult too soon** - Build confidence first
- **No guidance** - Provide scaffolding for beginners
- **Grading pressure** - CTFs should be low-stakes initially
- **Isolation** - Encourage collaboration and discussion
- **Tool overload** - Introduce tools gradually
- **Ignoring frustration** - Check in with struggling students

Resource Links

Platforms:

- CyberLeague: ctf.cyberleague.at
- picoCTF: picoctf.org
- TryHackMe: tryhackme.com
- HackTheBox: hackthebox.com

Tools:

- CyberChef: gchq.github.io/CyberChef
- Ghidra: ghidra-sre.org
- Burp Suite: portswigger.net/burp

Learning Resources:

- CTF Field Guide: trailofbits.github.io/ctf
- CTF101: ctf101.org

Q&A and Discussion

Questions?

💬 Let’s discuss!

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